EModel-Seek: Sequence Identification for Bottom-Up Cryo-EM Structural Discovery via Contrastive Learning

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Abstract

For decades, cryo-electron microscopy (cryo-EM) research has largely followed a top-down paradigm, in which a biological target or sequence is selected before its structure is determined. Emerging methods such as CryoSeek are advancing a bottom-up paradigm: first visualizing macromolecular assemblies by electron microscopy and then discovering their molecular identities from the observed densities. This shift calls for reliable recovery of sequence information from cryo-EM maps without prior knowledge of protein identity, making accurate residue-type discrimination a critical requirement. Here, we present EMXPlus, an enhanced deep learning framework that extends our previous EModelX pipeline by incorporating supervised contrastive learning to improve residue-type discrimination and sequence recovery. By explicitly shaping the latent embedding space to separate features of different amino acid types, EMXPlus achieves markedly sharper class boundaries and stronger generalization across varying resolutions. On a benchmark of 99 experimental cryo-EM maps at 2–4 Å resolution, EMXPlus attains 55% residue-type prediction accuracy, an absolute improvement of 7% over EModelX. Beyond residue-level prediction, EMXPlus integrates a sequence identification module that enables direct inference of protein sequences from cryo-EM maps without relying on any prior structural or sequence information. Querying against the UniProt database, EMXPlus achieves a mean sequence identity of 0.84, substantially outperforming findMySequence (0.56) while operating independently of pre-defined backbone models. Together, these advances establish EMXPlus as a general framework for sequence-resolved modeling of macromolecular complexes, representing a key step toward fully automated molecular identification and discovery directly from cryo-EM data.

1 Introduction

Cryo-electron microscopy (cryo-EM) has transformed structural biology by enabling the visualization of macromolecular assemblies at near-atomic resolution without the need for crystallization [1, 2, 3]. Accurate atomic model building from cryo-EM density maps is fundamental to elucidating molecular mechanisms, facilitating structure-guided drug discovery [4], and advancing structural genomics [5, 6]. Sequencing technologies have become well established, making protein sequence information available at unprecedented scale. Structural studies have largely followed a sequence-driven, top-down paradigm, in which a biological target is identified and its sequence established before its structure is determined.

Recent developments such as CryoSeek [7] have introduced a practical workflow for an emerging bottom-up paradigm of structure-guided biological discovery: first observing macromolecular assemblies and then determining their molecular identities. By enabling direct visualization of proteins and complexes in heterogeneous or native biological samples, cryo-EM offers opportunities to uncover previously uncharacterized molecular entities without requiring their identities to be known in advance. CryoSeek combines automated model building from cryo-EM density maps with Foldseek [8]–based structure matching, linking experimental densities to candidate proteins through searches against structural databases such as the AlphaFold Protein Structure Database (AFDB). Although structure-based comparison methods [9, 10] have proven powerful for homology detection and functional inference, their effectiveness depends on database coverage. Moreover, proteins with substantially different sequences can share similar folds [9, 11, 12], making structural similarity alone insufficient to establish the exact sequence of an observed entity. Candidate matches therefore often require residue-by-residue inspection and manual correction by experienced structural biologists [13, 14]. Such time-consuming, expertise-intensive curation remains a major bottleneck in large-scale cryo-EM map interpretation.

An alternative and potentially more general approach is to infer amino acid sequence information directly from cryo-EM density maps, thereby enabling sequence-based identification against comprehensive repositories such as UniProt [15]. This strategy could provide a direct link between density interpretation and molecular identity. Existing methods such as `findMySequence` [16] and EMSequenceFinder [17] estimate residue-type probabilities from pre-built backbone models and then perform probabilistic matching against sequence databases. However, these approaches rely on the availability of accurate backbone traces, a requirement that can be difficult to satisfy for unknown complexes, where chain segmentation and atomic connectivity remain uncertain.

Our previous work, EModelX [18], enables automated atomic model building from cryo-EM density maps given the corresponding protein sequences. Its deep learning framework predicts atomic distributions and residue-type identities from volumetric density data to support model construction. Building on EModelX, we have also developed methods for CryoSeek-related applications, including automated glycoprotein modeling [19] and handedness determination [20]. However, the identification of unknown proteins within the CryoSeek workflow still calls for a robust, automated framework that recovers sequence information from density maps and uses it to establish molecular identity. Accurate residue-type discrimination is central to this task, as errors in residue assignments can compromise both sequence recovery and subsequent database matching.

Here, we present EMXPlus, an enhanced deep learning framework that extends our previous EModelX pipeline by incorporating supervised contrastive learning [21] to improve residue-

type discrimination and sequence recovery. Supervised contrastive learning promotes inter-class separability and intra-class consistency by explicitly pulling together features of the same amino acid type while pushing apart those of different types [22], thereby refining the latent embedding space for amino acid discrimination. EMXPlus further incorporates a sequence identification module, forming a unified pipeline for *de novo* sequence recognition directly from cryo-EM density maps. By coupling voxel-level discriminative representation learning with downstream sequence-based searching, EMXPlus links cryo-EM density interpretation to molecular identification, providing an automated framework for discovering unknown proteins from experimental data.

2 Results

2.1 EMXPlus enhances amino-acid type discrimination in Cryo-EM maps.

EMXPlus extends the deep learning framework of EModelX by integrating supervised contrastive learning to enhance amino acid type prediction/discrimination. A schematic overview of the network is shown in Figure 1A. The model learns voxel-level representations of cryo-EM density maps by encouraging local features around atomic positions of the same residue type to be similar, while pushing apart those of different residue types to maximize discriminability.

Residue-type prediction performance was evaluated by comparing EMXPlus with the original EModelX on the benchmark dataset. As shown in **??**A, EMXPlus achieved an average accuracy of 0.54, exceeding the 0.48 obtained by EModelX. Moreover, **??**B demonstrates that EMXPlus consistently outperformed EModelX across nearly all test cases, indicating that the incorporation of supervised contrastive learning effectively enhances residue-type discrimination. We further analyzed the prediction accuracy with respect to cryo-EM map resolution (**??**C), revealing a strong positive correlation between amino acid prediction accuracy and map quality. Across the entire resolution range (2–4 Å), EMXPlus exhibited a clear trend of improved performance compared with EModelX.

2.2 Higher residue-type prediction accuracy contributes to improved sequence identification

To investigate whether improved residue-type classification enhances downstream sequence identification, we constructed atomic models from cryo-EM density maps and extracted predicted sequence fragments for database searching. The fragments were aligned against the UniProtKB using `phmmer`, and the sequence identity (SI) score between the retrieved entries and their corresponding ground-truth sequences was computed.

Using EModelX-generated models, the mean SI across all matched sequences was 0.78, whereas EMXPlus achieved a higher average of 0.81 (**??**B). Overall, 67% of EModelX-based searches were considered reliable, achieving both SI and reference coverage above 0.9, whereas EMXPlus reached 74%. Similarly, the proportion of exact matches (identity = 1.0) increased from 40% with EModelX to 45% with EMXPlus. At the protein level, mean identity scores were computed by averaging across all chains within each complex. For several representative cases, such as PDB IDs 7l48, 7apd, and 7rye, EModelX-based predictions failed to retrieve the correct sequences (identity < 0.8), while

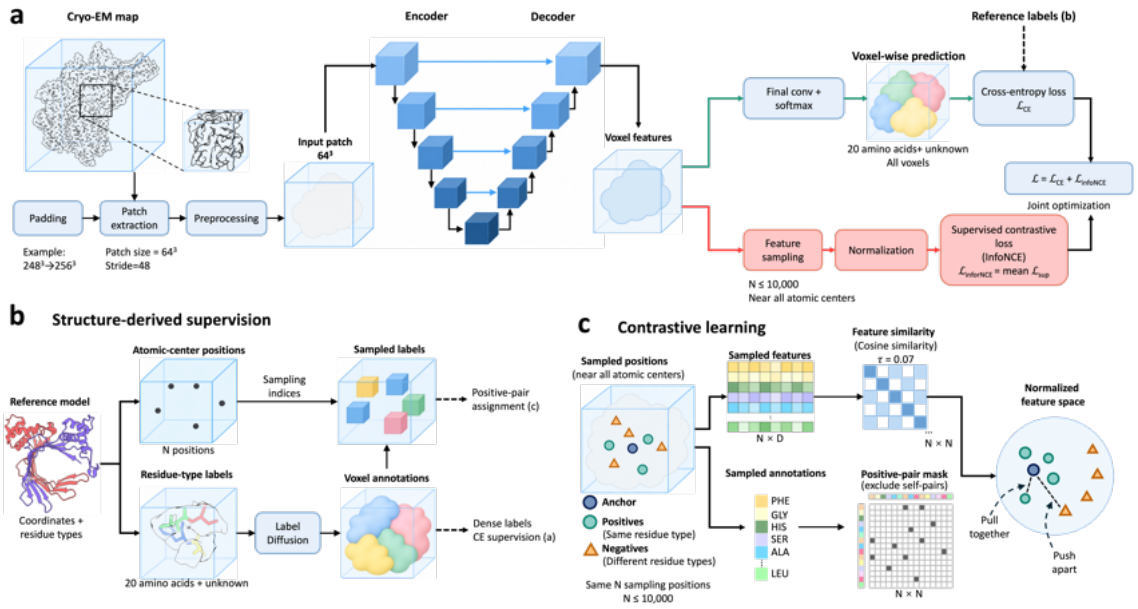


Figure 1: **EMXPlus integrates voxel-wise classification and supervised contrastive learning for amino acid type prediction. a,** Cryo-EM density patches are processed by a shared 3D U-Net with encoder–decoder skip connections. Voxel features support classification into 20 amino acid types and an unknown class, together with supervised contrastive learning on sampled features. Training jointly minimizes the cross-entropy loss $\mathcal{L}_{\mathrm{CE}}$ and the supervised contrastive loss $\mathcal{L}_{\mathrm{SupCon}}$, averaged over anchors. **b,** Structure-derived supervision. Reference atomic coordinates determine sampling locations at or near atomic centers, while residue-type assignments provide voxel labels aligned to the density-map grid. Dense labels supervise classification, and labels extracted at the sampled locations define positive-pair relationships. **c,** Voxel-level supervised contrastive learning. Features and labels are extracted at the same sampling locations. Normalized features yield pairwise cosine similarities, and labels define a positive-pair mask excluding self-pairs. For each anchor, other sampled voxels of the same amino acid type are positives; those of different types are negatives. The InfoNCE-based objective promotes within-type feature similarity and between-type separation. The feature-space diagram illustrates the optimization direction rather than an empirical embedding.

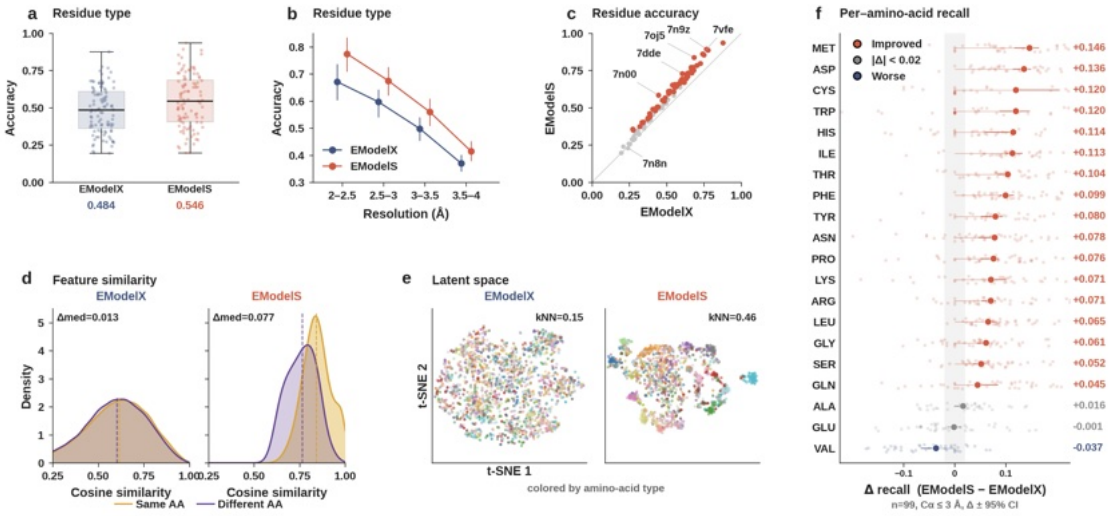


Figure 2: **Residue-type prediction and feature discrimination in EModelS.** EModelS, previously termed EMXPlus, was evaluated against EModelX on 99 cryo-electron microscopy (cryo-EM) maps. Deposited and predicted C_α positions were matched using a 3 Å spatial cutoff. Residue-type accuracy was calculated as the number of correctly typed matched deposited C_α positions divided by the total number of deposited C_α positions; unmatched deposited positions were counted as incorrect. **a.** Distribution of per-map residue-type prediction accuracy. Boxes indicate the interquartile range, centre lines indicate medians, whiskers extend to 1.5 times the interquartile range, and individual points represent maps. Numbers beneath the method labels indicate mean accuracies. **b.** Mean residue-type accuracy across map-resolution bins. The bins contain 12, 14, 35 and 38 maps, respectively. Points indicate bin means and error bars indicate 95% percentile bootstrap confidence intervals based on map-level resampling with 4,000 replicates. **c.** Paired comparison of per-map residue-type accuracy. The diagonal denotes equal performance, and points above the diagonal favour EModelS. Selected examples are annotated by Protein Data Bank (PDB) identifier. **d.** Distributions of cosine similarity between normalized residue features for pairs of the same amino-acid type (orange) or different amino-acid types (purple). Features were extracted from the embedding immediately before the classification head and sampled from labelled atomic positions in 80 grid patches. Same-type and different-type pairs were each sampled 40,000 times after pooling across patches. Dashed lines indicate distribution medians, and Δ_{med} denotes the median similarity for same-type pairs minus that for different-type pairs. **e.** Two-dimensional t-distributed stochastic neighbour embedding (t-SNE) projections of residue features, coloured by amino-acid type. Features were first reduced by principal component analysis to at most 30 dimensions; t-SNE was used for visualization only. At most 120 residues per amino-acid type were displayed. The annotated k-nearest-neighbour accuracy was computed with $k = 15$ using five-fold cross-validation in the original normalized feature space. **f.** Change in recall for each amino-acid type, calculated as EModelS minus EModelX. Recall was defined as the number of correctly classified deposited residues divided by the total number of deposited residues of that amino-acid type. Solid circles and numerical annotations indicate pooled residue-level changes across all maps; faint points show per-map changes. Horizontal bars indicate 95% percentile bootstrap confidence intervals for the per-map mean change, obtained by map-level resampling with 4,000 replicates. Red and blue denote positive and negative changes with absolute magnitude of at least 0.02, respectively; gray denotes $|\Delta| < 0.02$ for visualization and does not indicate statistical significance.

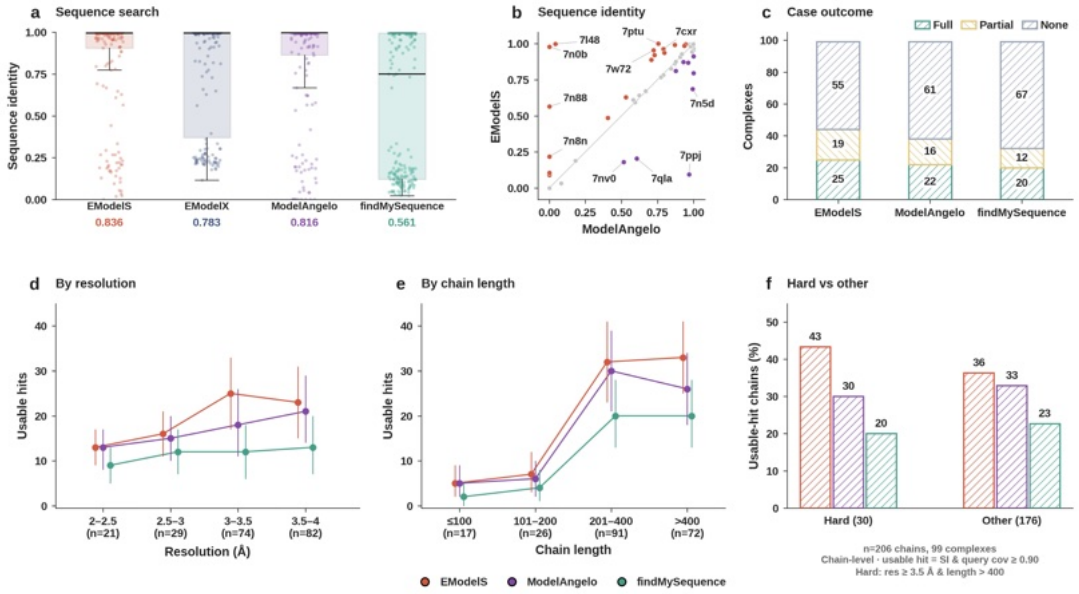


Figure 3: **Benchmarking protein sequence identification from cryo-EM maps.** Sequence identification was evaluated on the same set of 206 chains, each containing at least 30 residues, from 99 protein complexes. For each chain, local alignments were generated using EMBOSS `water`. Sequence identity was then recomputed by custom post-processing as the number of identical aligned residue pairs divided by the full reference-sequence length, and is therefore referred to as reference-normalized sequence identity (SI). Chains with no qualifying hit were assigned SI = 0 and retained in the common denominator. Query coverage was calculated as the number of aligned query residues divided by the query-sequence length. **a.** Distribution of per-chain reference-normalized SI for EModelS (red), EModelX (blue), ModelAngelo (purple) and `findMySequence` (teal). Boxes indicate the interquartile range, centre lines indicate medians, whiskers extend to 1.5 times the interquartile range, and individual points represent chains. Numbers beneath the method labels indicate mean SI. **b.** Comparison of per-complex mean SI for EModelS and ModelAngelo. For each complex, the mean was calculated across its evaluated chains after missing method-specific results had been assigned SI = 0. The diagonal denotes equal performance. Red and purple points indicate an EModelS or ModelAngelo advantage of at least 0.05, respectively; gray points indicate an absolute between-method difference below 0.05. Selected examples are annotated by PDB identifier. **c.** Complex-level identification outcomes for EModelS, ModelAngelo and `findMySequence`. A usable hit was defined by SI $\geq$ 0.90 and query coverage $\geq$ 0.90. Complexes were classified as Full when all evaluated chains had usable hits, Partial when only a subset had usable hits, and None when no evaluated chain had a usable hit. Numbers within the bars indicate the numbers of complexes in each category. **d,e.** Observed numbers of chains with usable hits, stratified by map resolution (**d**) and chain length (**e**). Error bars indicate 95% percentile bootstrap confidence intervals obtained by resampling chain-level indicators with 4,000 replicates. Parenthetical values indicate the numbers of evaluated chains in each bin. **f.** Percentage of chains with usable hits in the challenging subset, defined by map resolution $\geq$ 3.5 Å and chain length $>$ 400 residues (n = 30), and among all remaining chains (n = 176). Percentages were calculated as the number of usable-hit chains divided by the total number of evaluated chains within each subset. Values above the bars indicate percentages. Method colours are consistent across panels.

EMXPlus successfully identified the correct ones (identity > 0.9). Interestingly, these three cases have the same resolution of 3.9 Å, and the amino acid prediction accuracy improved by only about 1% with EMXPlus compared with EModelX, highlighting how even marginal gains in residue-type accuracy can lead to substantial improvements in sequence identification for low-resolution maps.

We further examined how sequence identity varied with amino acid prediction accuracy and map resolution. As shown in **??**D, although the distribution of SI values is somewhat polarized, a clear positive correlation exists between SI and amino acid prediction accuracy. This indicates that, while certain cases with low residue-type accuracy (AA accuracy < 0.4) can still achieve correct sequence retrieval (SI > 0.9) due to sequence-specific characteristics, higher amino acid prediction accuracy generally reduces the likelihood of sequence identification failures and increases positive predictive power. Notably, the regression line of EMXPlus (green) shifts upward and rightward compared with EModelX (yellow), reflecting both improved overall residue-type accuracy and enhanced sequence identification especially at lower accuracy levels.

In **??**E, we observe that EMXPlus yields markedly higher SI values for lower-resolution maps, with a gentler regression slope, indicating reduced sensitivity to resolution variations. We also analyzed the performance of sequence identification across different chain lengths (**??**F). EMXPlus consistently outperformed EModelX, particularly for shorter chains, where the correct identification of a few additional residues has a proportionally greater impact.

Taken together, these results demonstrate that incorporating supervised contrastive learning not only improves residue-type prediction but also enhances the robustness and reliability of downstream sequence identification, especially for difficult cases with low-resolution cryo-EM data or short-chain proteins.

2.3 EMXPlus demonstrates improved performance in protein sequence identification

We further compared the performance of EMXPlus and `findMySequence` in protein sequence identification. Since `findMySequence` performs sequence searching using multiple sequence alignments (MSAs) derived from its predicted residue-type probability distributions, we evaluated EMXPlus under the same MSA-based protocol using `hmmsearch`, querying the UniProtKB in both cases. As shown in **??**B, EMXPlus achieved an average SI of 0.84 using MSA-based searching, slightly higher than the 0.82 obtained through direct sequence-based searching. Both approaches substantially outperformed `findMySequence`, which achieved only 0.56 SI under identical benchmarking conditions.

Unlike `findMySequence`, which requires a pre-defined protein backbone as input, EMXPlus directly infers sequence information from cryo-EM density maps without relying on prior structural knowledge. This enables EMXPlus to function as a unified framework for both structural modeling and *de novo* sequence identification. Across the benchmark set, 76% of EMXPlus-based searches achieved both SI and reference coverage above 0.9, compared to 44% for `findMySequence`. Correlation analyses (**??**E–F) further show that EMXPlus consistently outperforms `findMySequence` across all resolutions and chain lengths, with the advantage particularly evident for short chains, where `findMySequence` often fails due to insufficient structural context. These results highlight the robustness and general applicability of EMXPlus in sequence

recovery from cryo-EM data.

Although MSA-based searching provides only marginal overall improvement compared with direct sequence searching, it tends to perform better for low-resolution maps and longer proteins. This is likely because MSAs capture evolutionary constraints that compensate for local residue-type prediction errors. In contrast, for high-resolution maps where residue-level accuracy is already high, MSA incorporation increases computational cost without clear benefit. For short chains, MSAs may even introduce stochastic variations that destabilize search results.

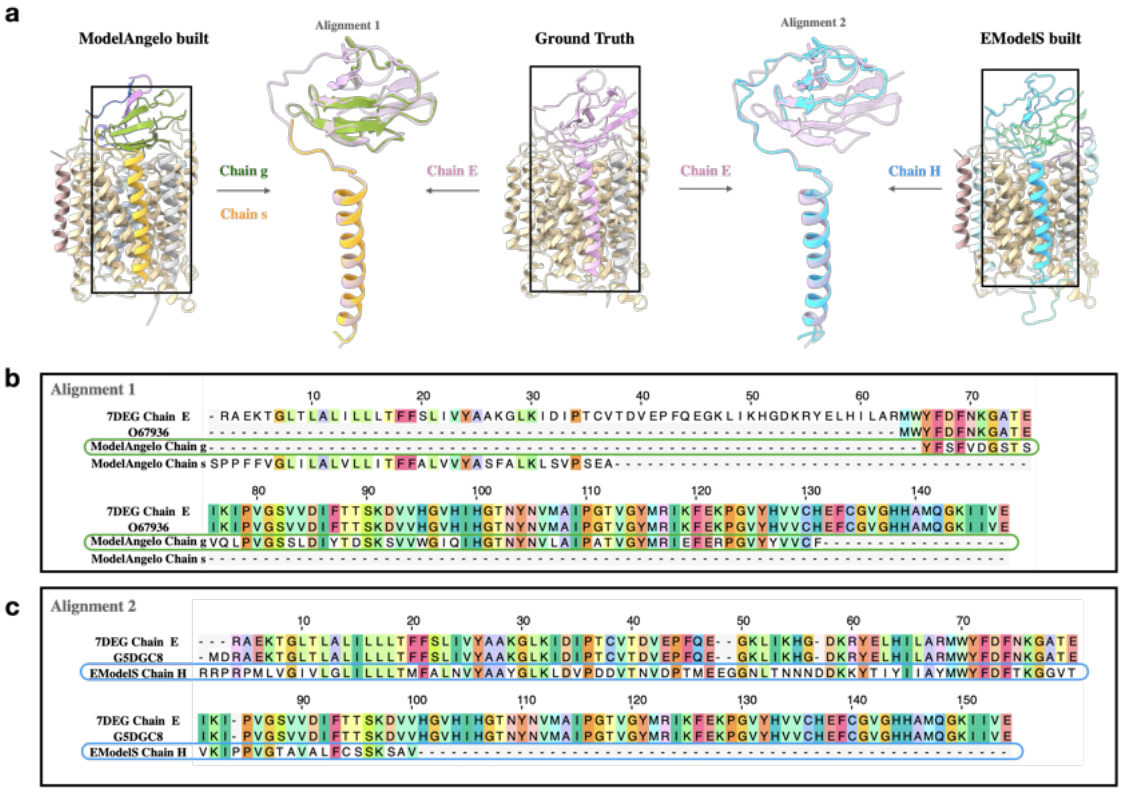


Figure 4: **NEW CASE A.** 1 **B.** 2 **C.** 3 **D.** 4 **E.** 5

2.4 Higher residue-type prediction accuracy contributes to improved structure building

To examine how improvements in amino acid distribution prediction translate into the ultimate goal of cryo-EM map interpretation, which is the accurate structural discovery, we analyzed the case of the sodium-driven bicarbonate exchanger NDCBE[23] (Figure 5). In this example, EMXPlus achieved an amino acid type prediction accuracy of 0.44, compared to 0.39 for EModelX. When the sequence information was provided for full atomic model building, the model generated by EModelX exhibited incorrect regional chain assignments during sequence alignment, whereas

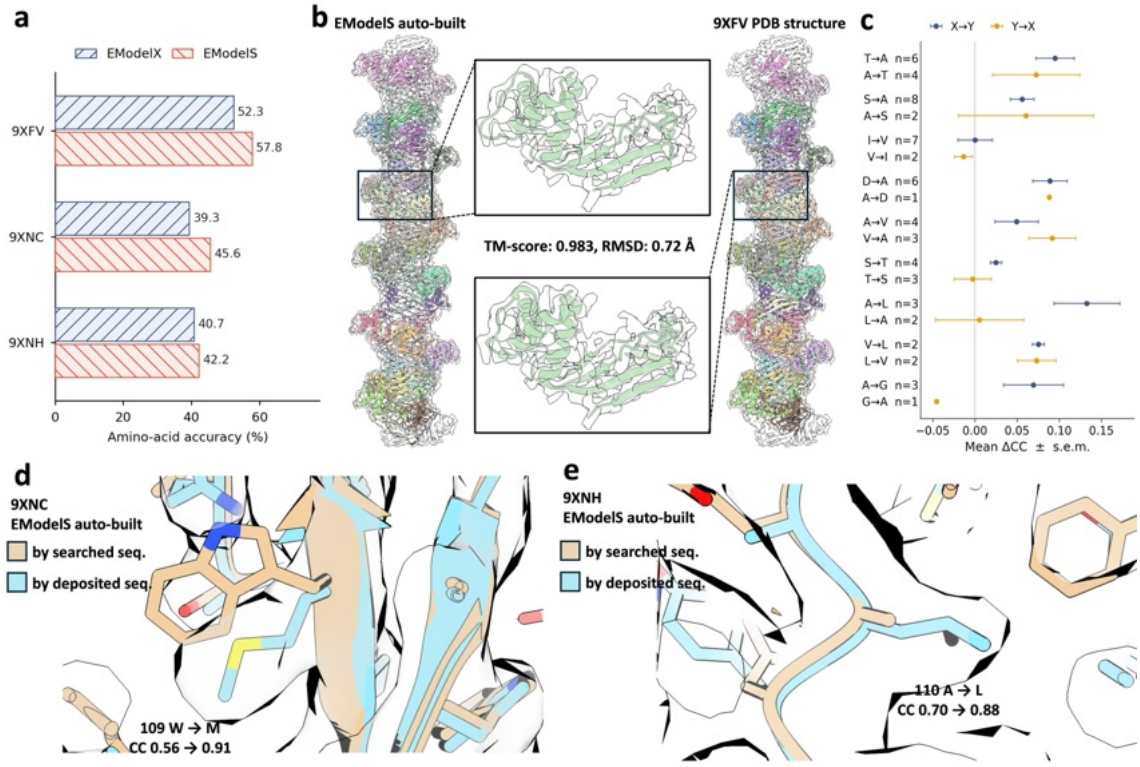


Figure 5: **Comparison of the predicted models of the sodium-driven bicarbonate exchanger NDCBE (SLC4A8) with the ground-truth PDB structure. A.**The upper panel shows the ground-truth structure of the protein together with the spatial distribution of C_α atoms whose residue types were correctly predicted by EModelX or EMXPlus. Red dots indicate C_α atoms correctly classified by EModelX, whereas blue dots mark those correctly identified by EMXPlus. The lower panels display the models built by EModelX and EMXPlus, respectively, using the same Cryo-EM map and sequence of NDCBE as input. For each model, C_α atoms with correctly predicted residue types are highlighted as dots in the corresponding structures. All models are fitted into the experimental cryo-EM density map and colored by chain. Two key regions exhibiting improved local modeling by EMXPlus compared to EModelX are indicated by boxes. **B.** Magnified view of region 1 near one end of the chains, where accurate prediction of local amino acid distributions enabled correct chain connections and consequently improved global chain assignment. **C.** Magnified view of region 2 near the opposite end of the chains, where accurate prediction of amino acid distributions similarly resulted in correct chain connectivity, contributing greatly to overall chain assignment accuracy.

EMXPlus successfully resolved these errors. Further inspection revealed that these corrections primarily originated from two key regions where EMXPlus demonstrated superior performance in local structure building.

In Figure 5, region 1 (solid box) highlights an area where the improved amino acid classification enabled EMXPlus to correctly assign chain A (pink), while EModelX incorrectly classified the same segment as chain B (green). Additionally, accurate residue-type predictions at the critical site within the dashed box allowed EMXPlus to establish the correct chain connection that EModelX failed to build. In region 2, a similar improvement was observed: higher local residue-type accuracy led to more precise modeling of the chain junctions. Since NDCBE contains two chains (A and B) with identical sequences, accurate modeling near chain termini is particularly crucial. Both highlighted regions are located near the chain ends, where EMXPlus's corrections not only improved the local geometry but also facilitated better continuation of the downstream segments, yielding a substantially improved overall model. It is also likely that the enhanced residue-type predictions across other regions collectively contributed to the improvement. This case illustrates how a 5% increase in residue-type prediction accuracy can lead to significant gains in the quality and correctness of the predicted atomic model. This observation also provides valuable insight into scenarios where the sequence information of a protein is unknown. Improved amino acid type distribution prediction simultaneously enhances both sequence identification and atomic model building, with the improvement in sequence identification further reinforcing the overall modeling accuracy. Consequently, enhancing amino acid prediction accuracy proves to be highly effective, particularly in de novo structure determination tasks.

3 Discussion

Traditional sequence identification methods typically rely on pre-determined backbone atomic coordinates, assuming that the positions and boundaries of protein chains are known in advance. In practical structural modeling tasks, however, such chain segmentation often depends on labor-intensive manual interpretation of cryo-EM density maps, where segmentation accuracy directly determines the reliability of subsequent sequence identification. Consequently, these conventional approaches function primarily as auxiliary tools rather than fully automated solutions for protein structure modeling. Other sequence identification strategies assume that the protein sequence is known a priori, relying on sequence alignment or profile-based homology analysis [24]. In summary, achieving reliable sequence identification without prior sequence information remains a major unsolved challenge in *de novo* structural biology.

For *de novo* protein identification, an alternative class of methods performs structure-based homology searches by identifying atomic-level similarities to known models [8, 10]. Although these approaches have achieved remarkable success in detecting structural homologs, structural similarity alone rarely provides reliable sequence-level correspondence. Even proteins with nearly identical folds can exhibit very low sequence identity, and their identification still requires residue-by-residue manual validation. As a result, structure-based workflows still depend on expert intervention for residue assignment and register-shift correction [13, 14], limiting their scalability and automation potential.

In contrast, EMXPlus extracts chain information directly from atomic models predicted from cryo-EM density maps and subsequently identifies their corresponding sequences without any prior

structural or sequence knowledge. This capability is particularly valuable in realistic modeling scenarios where the molecular composition is unknown or heterogeneous. Even when the identified sequences are not perfectly accurate, the candidate sequences predicted by EMXPlus provide valuable biochemical cues that can guide iterative model refinement. These predicted sequences often share similar biochemical or structural characteristics—such as belonging to the same protein family or deriving from related biological contexts—thereby offering meaningful contextual information for downstream analysis, including homologous sequence searches [25]. Moreover, correctly identifying even a single high-confidence chain can facilitate the recognition of neighboring regions and enable iterative refinement of preliminary assignments, for example by correcting register shifts using identified sequences as in checkMySequence [14]. Although such post-processing steps were not explicitly implemented in this study, EMXPlus provides a robust foundation that can be seamlessly integrated into existing cryo-EM modeling pipelines for comprehensive protein identification and characterization.

Compared with existing sequence identification approaches, EMXPlus achieves substantially higher sequence recognition accuracy. Once the protein sequences are correctly identified, atomic models can be reconstructed from cryo-EM data with accuracy comparable to cases where the sequences are known a priori. This finding highlights that accurate sequence identification is not merely a byproduct of structure modeling but a critical determinant of modeling performance for previously uncharacterized proteins.

The improvement in sequence identification fundamentally arises from enhanced residue-type prediction. Our results demonstrate that even small corrections in residue-type classification at a few key positions can lead to significant improvements in overall sequence recovery accuracy. This observation underscores the importance of employing advanced learning paradigms such as supervised contrastive learning to enhance residue-type discriminability. In our current benchmark, nearly half of the Cα positions remain misclassified, revealing considerable room for further improvement. EMXPlus incorporates contrastive learning at the voxel level, balancing predictive precision with computational efficiency. Recent advances in contrastive representation learning for three-dimensional data [26, 27, 28, 29] provide valuable inspiration; for example, spatially aware sliding-window sampling could enable the network to capture complete residue patterns across larger spatial contexts, potentially leading to superior amino acid classification performance.

Future research directions include estimating residue-level confidence based on statistical analysis of correctly predicted Cα positions and incorporating these confidence scores into sequence identification. Integrating such probabilistic confidence weighting could allow alignment algorithms to prioritize high-certainty regions, thereby enhancing both robustness and accuracy in protein sequence recovery from cryo-EM density maps.

4 Methods

4.1 Datasets

High-resolution single-particle cryo-EM density maps were obtained from the Electron Microscopy Data Bank (EMDB) [30], each associated with a fitted atomic model in the Protein Data Bank (PDB).

For network training and validation, 6,082 and 89 maps released after January 2018 with reported

resolutions between 1 Å and 6 Å were selected after removing entries with high sequence similarity. For independent testing, 99 single-particle maps released after January 2021 with resolutions between 2 Å and 4 Å were used, consistent with the benchmark employed in EModelX.

For network input, each voxel in the experimental density map was labeled with its ground-truth amino acid type (one-hot encoded over the 20 standard residues) and a multi-dimensional set of binary indicators denoting whether it contained backbone atoms, C_α atoms, other heavy atoms, or was proximal to a C_α atom, as derived from the corresponding PDB structure. All density maps were preprocessed by removing low-density voxels, clipping extreme intensity values, and normalizing by the maximum voxel intensity. From each processed map, cubic subvolumes of $64 \times 64 \times 64$ voxels were extracted with a stride of 48, and the density and label volumes were concatenated and inputted into our network.

4.2 Contrastive learning framework for amino acid prediction

Accurate residue-type prediction is critical for reliable sequence identification from cryo-EM maps. To improve discriminability among amino acid features, EMXPlus extends the U-Net [31] backbone of EModelX by augmenting voxel-wise classification with a supervised contrastive learning objective. This framework jointly optimizes dense voxel-wise classification and representation-level feature separation, providing complementary supervision signals for both global and local learning.

Voxel-wise classification. A cross-entropy (CE) loss $\mathcal{L}_{\text{CE}}$ is computed over all voxels to provide dense supervision:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{21} y_{i,c}\,\log(\hat{y}_{i,c}),$$

where each voxel is labeled by a one-hot vector $y_i \in \{0,1\}^{21}$ representing 20 amino acids and one unknown class. This loss enforces voxel-level accuracy across the full density map.

Supervised contrastive loss. To enhance class-wise feature separability, we further employ a supervised contrastive loss applied to a subset of voxels located at or near atomic centers. Let S denote the sampled voxel subset. For an anchor voxel $i \in S$ with embedding z_i, the set $P(i) \subseteq S$ contains other voxels of the same residue type, and $A(i) = S \setminus \{i\}$ denotes all remaining samples. The per-anchor loss is defined as:

$$\mathcal{L}_{\text{sup}}(i) = -\frac{1}{|P(i)|}\sum_{p\in P(i)} \log\frac{\exp\big(\text{sim}(z_i, z_p)/\tau\big)}{\sum_{a\in A(i)}\exp\big(\text{sim}(z_i, z_a)/\tau\big)}, \quad \text{sim}(u,v) = \frac{\langle u, v\rangle}{\|u\|\|v\|}.$$

This formulation follows the InfoNCE principle [32], which minimizes the cross-entropy between the distributions of positive and negative pairs in the embedding space. The exponential normalization across all negatives effectively defines a softmax contrastive classifier, rendering the objective equivalent to maximizing the mutual information between embeddings of the same residue type while minimizing that across different types. A temperature parameter τ controls the sharpness of the softmax distribution: lower values increase contrast but risk gradient instability, whereas higher values reduce discriminative power [21]. Here we set $\tau = 0.07$, which offered a balance between training stability and inter-class margin in our validation experiments. Averaging over all anchors yields the batch-level contrastive loss $\overline{\mathcal{L}_{\text{sup}}}$.

Overall objective. The complete training objective combines both terms:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \overline{\mathcal{L}_{\text{sup}}}.$$

The CE component enforces dense voxel-level classification accuracy, while the contrastive term regularizes the embedding space to improve residue-type discriminability. Gradients from both losses are jointly back-propagated through the shared U-Net backbone, allowing EMXPlus to learn both accurate voxel-level classifications and residue-type–aware feature representations in a unified training process.

Sampling strategy. We evaluated several voxel sampling strategies to assess their influence on amino acid type prediction. Sampling all voxels or only Cα-neighboring voxels both led to 5% lower accuracy compared with sampling voxels located at or near all atomic centers. Although contrastive learning is generally more effective with larger sample sets [33, 34, 21], in our voxel-based setting, simply increasing the sampling scope introduced substantial redundancy and noise. The former strategy, which samples all voxels, included a large proportion of low-density or background regions that do not correspond to meaningful atomic structures, thereby degrading the quality of learned representations and slowing convergence. Conversely, restricting sampling to only Cα-neighboring voxels excluded explicit side-chain features critical for residue-type discrimination. Sampling voxels near all atomic centers thus achieves an effective compromise, retaining structural relevance while preserving sufficient sample diversity.

Ablation on loss design. We also trained a variant using only $\mathcal{L}_{\text{sup}}$ for representation learning and applied $\mathcal{L}_{\text{CE}}$ solely at the final classification head. This setup isolates the contribution of contrastive learning and achieved accuracy comparable to the joint-training configuration, indicating that the model can learn intrinsic residue-type separability purely from voxel features. This observation supports our interpretation that contrastive learning enables the model to form intrinsically discriminative feature clusters corresponding to distinct amino acid types, even in the absence of explicit class supervision.

4.3 Sequence Identification from Cryo-EM Maps

We processed the 99 cryo-EM density maps in the test dataset using the U-Net backbone of EMXPlus to obtain voxel-wise amino acid probability distributions. These predictions were subsequently fed into the downstream modules of EModelX to generate complete protein models without using any prior sequence information. For each predicted model, chains longer than 30 residues were extracted, as shorter fragments are typically insufficient for reliable sequence alignment.

The sequences of these extracted chains, together with their reversed counterparts, were searched against the UniProtKB database using `phmmer` [35], which runs the HMMER sequence search engine [36].

In parallel, residue-type log-probabilities at the Cα positions were converted into profile hidden Markov models (HMMs) in both forward and reverse directions for each chain. Each profile HMM was parameterized from the per-position amino acid log-probabilities, with match-state transition probabilities derived from residue-level confidence scores ($m \to m = \max(\text{confidence} - \delta, \gamma)$, with $\delta = 0.05$ and $\gamma = 0.5$). The resulting HMMER3-formatted profiles were then queried

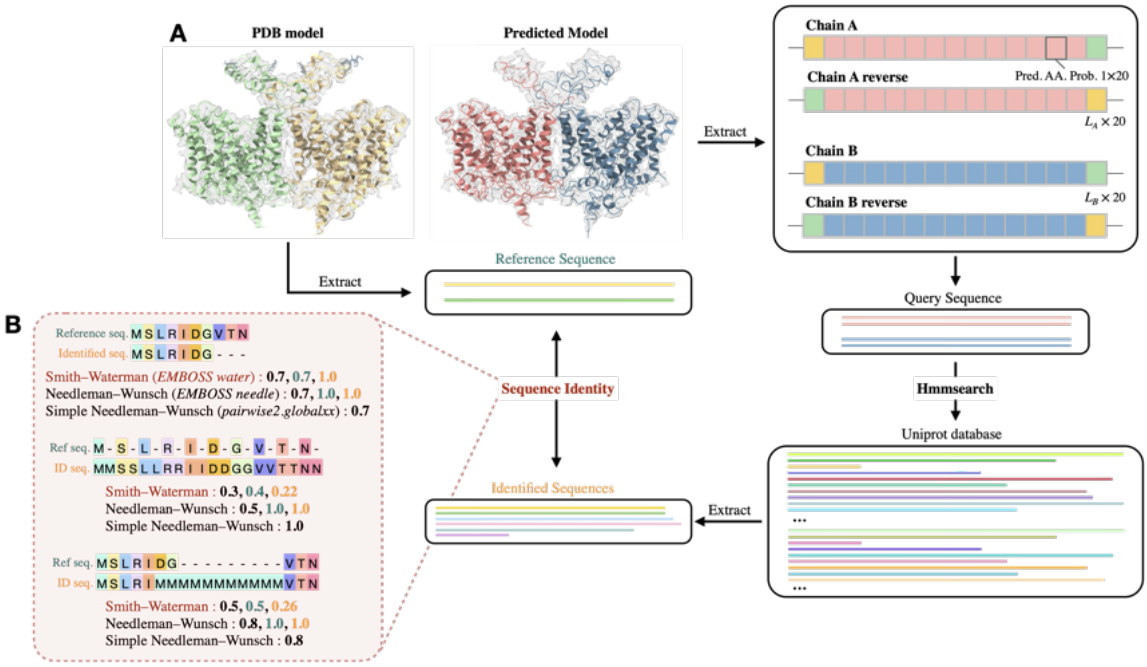


Figure 6: **A.** Overview of the proposed pipeline for identifying protein sequences from automatically built models. Protein chains and their corresponding sequences are shown in matching colors, with distinct sequences represented as different colored lines for clarity. **B.** Example comparisons of identified sequences against reference sequences using different alignment algorithms. Our method employs a modified Smith–Waterman scheme (red) to compute sequence identity (black), reference sequence coverage (green), and identified sequence coverage (yellow).

against UniProtKB using `hmmsearch` implemented in the HMMER3 package [37] to obtain profile-HMM–based identification results.

All candidate hits were filtered using a stringent threshold of $E < 10^{-10}$ to ensure high-confidence matches. For each chain, the top-ranked hit by E-value was retained as its identified sequence, while for each protein, the top ten hits across all chains were also recorded to capture alternative plausible matches.

4.4 Benchmarking Residue-Type and Sequence Identification

To compare residue-type prediction accuracy, EMXPlus and the original EModelX were evaluated on the same test set. For each model, the predicted amino acid type at every Cα position was extracted and compared with the ground-truth labels to compute residue-type accuracy. To benchmark sequence identification, we compared EMXPlus with both EModelX and `findMySequence`. For `findMySequence`, the reference backbone structures from the corresponding PDB files were provided as input to generate the identified sequences, whereas EMXPlus and EModelX directly produced identified sequences from raw cryo-EM density map inputs, by automatic model predicting procedures.

Sequence identification was assessed using a pairwise alignment procedure derived from the EMBOSS[38] `water` algorithm (Smith–Waterman local alignment[39]), further adapted through customized post-processing to meet our evaluation criteria. Rather than relying on the default alignment scores, we explicitly parsed the alignment strings and recomputed residue-level counts. A residue pair was considered a match only if both residues were non-gap characters and identical, while aligned positions on each side were counted whenever a residue was non-gap regardless of equality. This definition eliminates artifacts from discontinuous or trivially short local matches and decouples alignment quality from terminal penalties that typically bias global alignment schemes. As illustrated in Figure 6B, our approach provides a more stringent and coverage-sensitive evaluation compared with two commonly used Needleman–Wunsch–based[40] implementations (EMBOSS `needle` and `pairwise2.globalxx in Biopython`).

For a protein consisting of a chain set $\mathcal{C}$ (excluding chains shorter than 30 residues), let L_c denote the length of the reference sequence for chain c, M_c the number of exact residue matches (non-gap and identical), R_c the number of aligned residues on the reference side (non-gap positions in the reference), and Q_c the number of aligned residues on the query side (non-gap positions in the identified sequence) with query length $|q_c|$. We define the aggregated per-protein metrics as

$$\text{Score} = \frac{\sum_{c\in\mathcal{C}} M_c}{\sum_{c\in\mathcal{C}} L_c}, \qquad \text{RefCov} = \frac{\sum_{c\in\mathcal{C}} R_c}{\sum_{c\in\mathcal{C}} L_c}, \qquad \text{QryCov} = \frac{\sum_{c\in\mathcal{C}} Q_c}{\sum_{c\in\mathcal{C}} |q_c|}.$$

Here, Score measures the proportion of reference residues that are exactly recovered, while RefCov and QryCov quantify the completeness of the alignment from the reference and query perspectives, respectively. Unlike the alignment-length–normalized identity reported by default in EMBOSS or the purely global alignment score, our reference-length normalization penalizes fragmentary or overfitted local matches and prevents short, high-identity regions from dominating the overall evaluation. For each reference sequence, the highest identity score among all retrieved candidate sequences was recorded for each method to ensure a fair comparison. For completeness, we also report auxiliary statistics such as alignment length, alignment-normalized identity, gap fraction, and

the raw alignment score; however, our comparative analyses and rankings are primarily based on the reference-normalized Score, RefCov, and QryCov. A comprehensive summary of these metrics for all test cases across different methods is provided in Supplementary Table S1.

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